project name: Devali sales data analysis

> This is dataset belongs to sales & marketing data analysis catagory
>
> Dataset is made by perplexty AI for real-Market Simulation and analysis
>
> ## Now the Problems are:
>
> > - Where majority of products are sold?
> > - Most Sold Product ID?
> > - which are the ideal customers to target?
> > - their Age, Gender, Occupation.

Libraries Loading

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading dataset

In [2]:
```
df=pd.read_csv("Diwali Sales Data.csv",encoding="unicode_escape")
```

step 1: Understanding data

Data Shape and Columns

In [3]:
```
df.shape
```

Out[3]:
```
(11251, 15)
```

In [4]:
```
df.columns
```

Out[4]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount', 'Status', 'unnamed1'],
      dtype='str')
```

first 10 columns Info and Description

In [5]:
```
df.head(10)
```

Out[5]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Mah
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar
3	1001425	Sudevi	P00237842	M	0-17	16	0	K
4	1000588	Joni	P00057942	M	26-35	28	1	
5	1000588	Joni	P00057942	M	26-35	28	1	I
6	1001132	Balk	P00018042	F	18-25	25	1	Uttar
7	1002092	Shivangi	P00273442	F	55+	61	0	Mah
8	1003224	Kushal	P00205642	M	26-35	35	0	Uttar
9	1003650	Ginny	P00031142	F	26-35	26	1	Andhra

In [6]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   User_ID           11251 non-null  int64
 1   Cust_name         11251 non-null  str
 2   Product_ID        11251 non-null  str
 3   Gender            11251 non-null  str
 4   Age Group         11251 non-null  str
 5   Age               11251 non-null  int64
 6   Marital_Status    11251 non-null  int64
 7   State             11251 non-null  str
 8   Zone              11251 non-null  str
 9   Occupation        11251 non-null  str
 10  Product_Category  11251 non-null  str
 11  Orders            11251 non-null  int64
 12  Amount            11239 non-null  float64
 13  Status            0 non-null      float64
 14  unnamed1          0 non-null      float64
dtypes: float64(3), int64(4), str(8)
memory usage: 1.3 MB
```

In [7]:
```
df.describe
```

Out[7]:
```
<bound method NDFrame.describe of        User_ID    Cust_name Product_ID Gend
er Age Group  Age  Marital_Status  \
0      1002903    Sanskriti  P00125942      F     26-35   28               0
1      1000732       Kartik  P00110942      F     26-35   35               1
2      1001990        Bindu  P00118542      F     26-35   35               1
3      1001425       Sudevi  P00237842      M      0-17   16               0
4      1000588         Joni  P00057942      M     26-35   28               1
...        ...          ...        ...    ...       ...  ...             ...
11246  1000695      Manning  P00296942      M     18-25   19               1
11247  1004089  Reichenbach  P00171342      M     26-35   33               0
11248  1001209        Oshin  P00201342      F     36-45   40               0
11249  1004023       Noonan  P00059442      M     36-45   37               0
11250  1002744      Brumley  P00281742      F     18-25   19               0

                State      Zone       Occupation Product_Category  Orders  \
0         Maharashtra   Western       Healthcare             Auto       1
1      Andhra Pradesh  Southern             Govt             Auto       3
2       Uttar Pradesh   Central       Automobile             Auto       3
3           Karnataka  Southern     Construction             Auto       2
4             Gujarat   Western  Food Processing             Auto       2
...               ...       ...              ...              ...     ...
11246     Maharashtra   Western         Chemical           Office       4
11247         Haryana  Northern       Healthcare       Veterinary       3
11248  Madhya Pradesh   Central          Textile           Office       4
11249       Karnataka  Southern      Agriculture           Office       3
11250     Maharashtra   Western       Healthcare           Office       3

        Amount  Status  unnamed1
0      23952.0     NaN       NaN
1      23934.0     NaN       NaN
2      23924.0     NaN       NaN
3      23912.0     NaN       NaN
4      23877.0     NaN       NaN
...        ...     ...       ...
11246    370.0     NaN       NaN
11247    367.0     NaN       NaN
11248    213.0     NaN       NaN
11249    206.0     NaN       NaN
11250    188.0     NaN       NaN

[11251 rows x 15 columns]>
```

Step 4: DataSet Cleaning

data cleaning steps

```
1:fix missing values
2:removing duplicates
3:correcting datatype
4:cleaning objects (spaces, case issues)
```

applied steps

```
1:fix missing values
```

> Only deleted 2 coluumns which were filled with null and dropped 12 missing rows

> > > All data Format is correct as it should be
> > >
> > > - No duplicates
> > > - No format issues
> > > - No outliers

In [8]:
```
df.drop(['Status','unnamed1'],axis=1,inplace=True)
```

In [9]:
```
df.isnull().sum()
```

Out[9]:
```
User_ID              0
Cust_name            0
Product_ID           0
Gender               0
Age Group            0
Age                  0
Marital_Status       0
State                0
Zone                 0
Occupation           0
Product_Category     0
Orders               0
Amount              12
dtype: int64
```

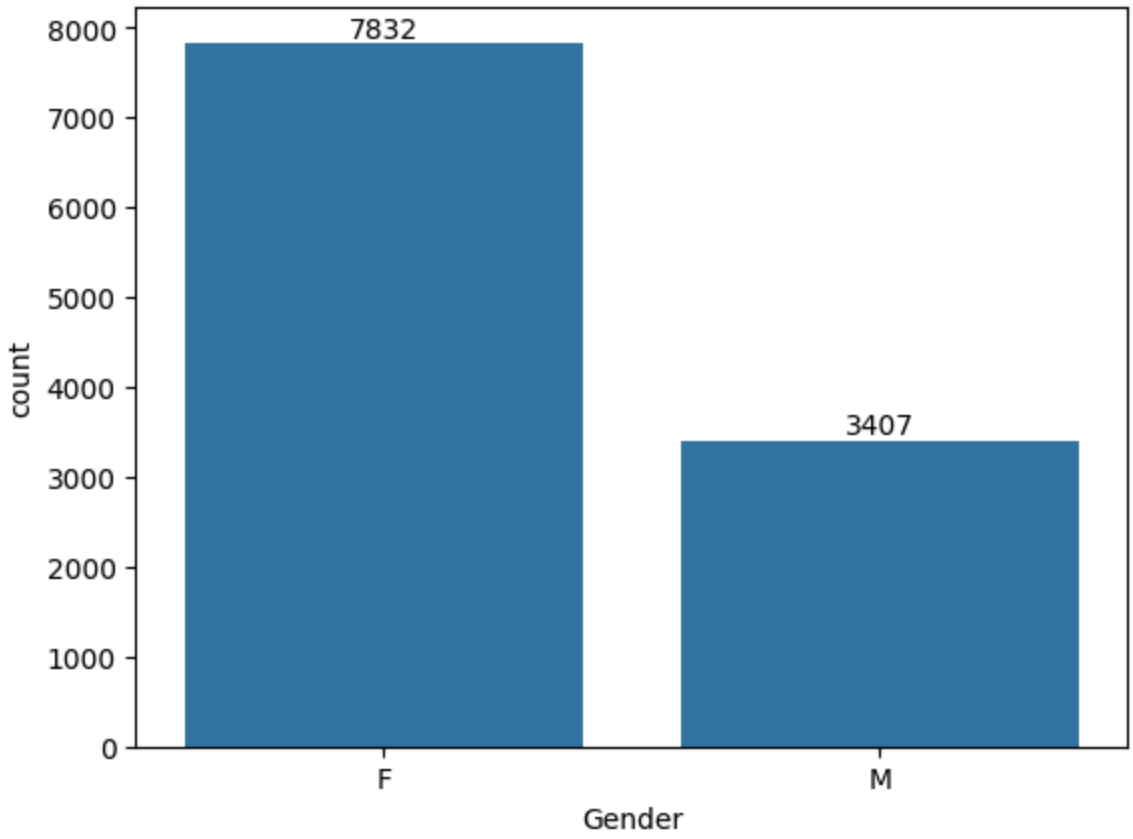
In [10]:
```
df.dropna(inplace=True)
```

In [11]:
```
df["Amount"]=df["Amount"].astype(int)
```

EDA 5.1: Step data visualization

> ## customers count by gender

In [12]:
```
graph=sns.countplot(x='Gender',data=df)
for bars in graph.containers:
    graph.bar_label(bars)
```



> ## result
>
> - Univariant analysis
>
> > - 7832 Females
> > - 3407 Males
>
> Takeaway from graph: **most buyers are females**

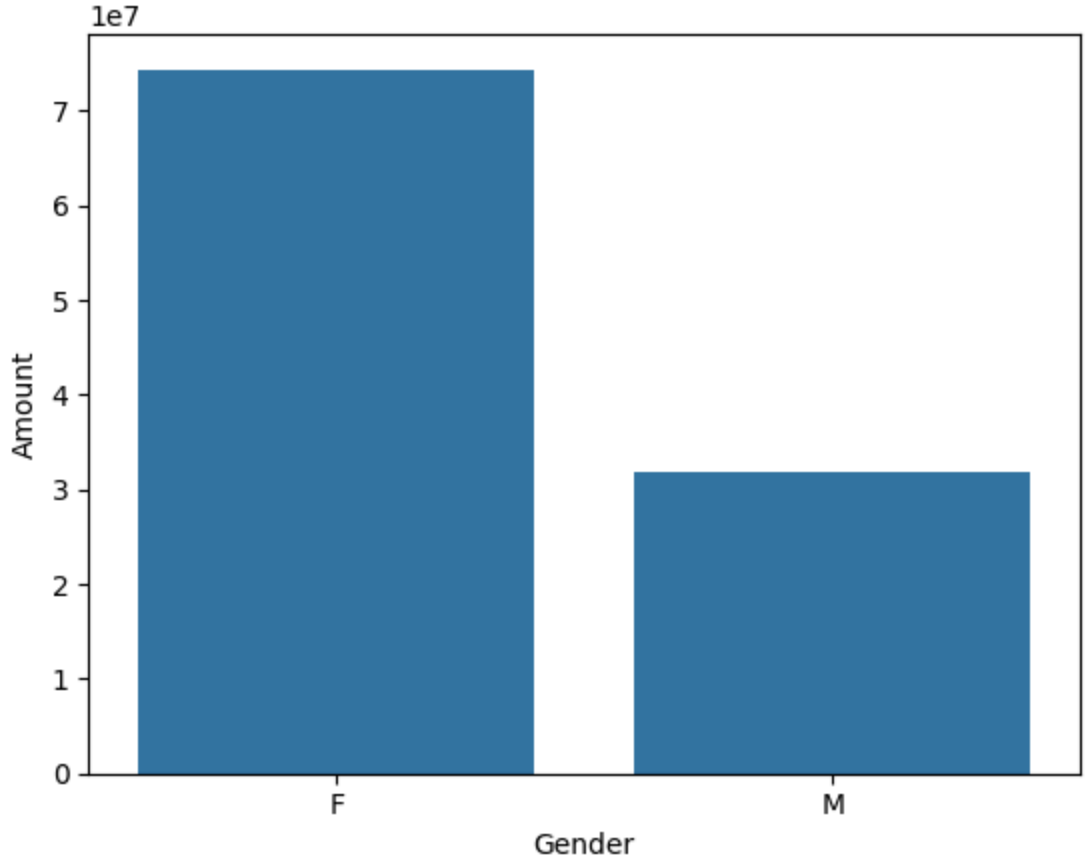
step 5.3 Univariate data analysis

> ## Grouping amount by gender

In [13]:
```
snsplot=df.groupby(['Gender'], as_index=False)['Amount'].sum().sort_values(by=
```

In [14]:
```
sns.barplot(x="Gender",y="Amount",data=snsplot)
```

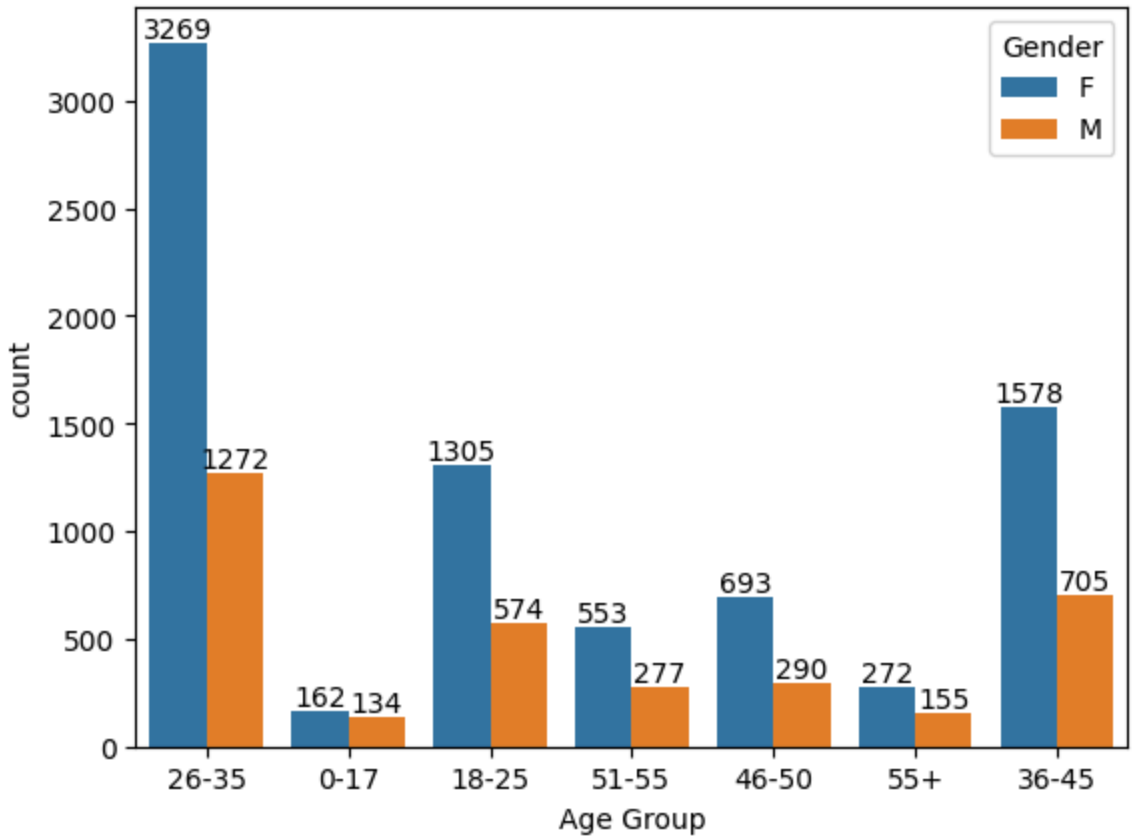
Out[14]:
```
<Axes: xlabel='Gender', ylabel='Amount'>
```



> ## Result of bivariant analysis
>
> > - Females: 7.3-e7.RS
> > - Males: 3.1-e7.RS
>
> Takeaway from graph: **Females spend more amount of money on shopping than males**

> # Plotting count of age group by Gender

In [15]:
```
graph=sns.countplot(data=df,x="Age Group",hue="Gender")
for bars in graph.containers:
    graph.bar_label(bars)
```

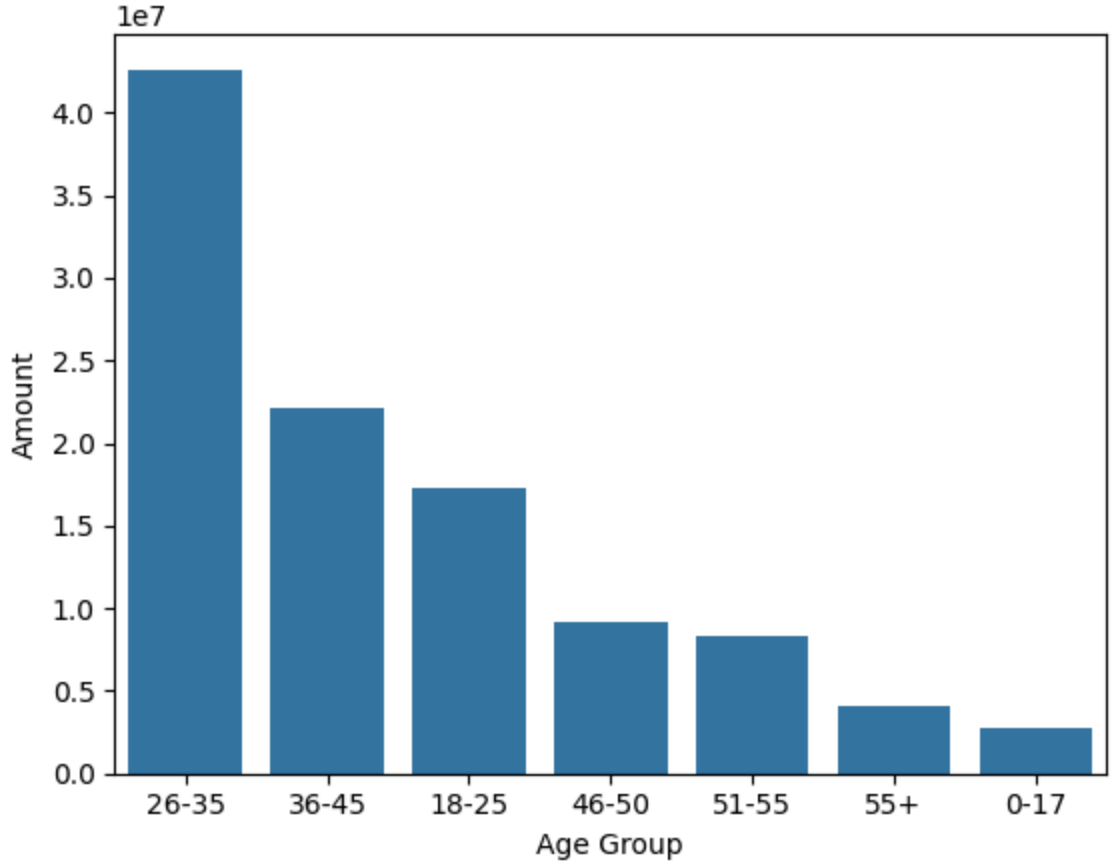


> ## Result of by bivariant analysis
>
> > takeaway from graph: Females of age group 26-35 are majority purchasing and age group 0-17 is the least purchasing group

> ## plotting of age group by amount

In [16]:
```
gOfSa=df.groupby(['Age Group'], as_index=False)['Amount'].sum().sort_values(by
sns.barplot(x='Age Group',y='Amount',data=gOfSa)
```

Out[16]:
```
<Axes: xlabel='Age Group', ylabel='Amount'>
```



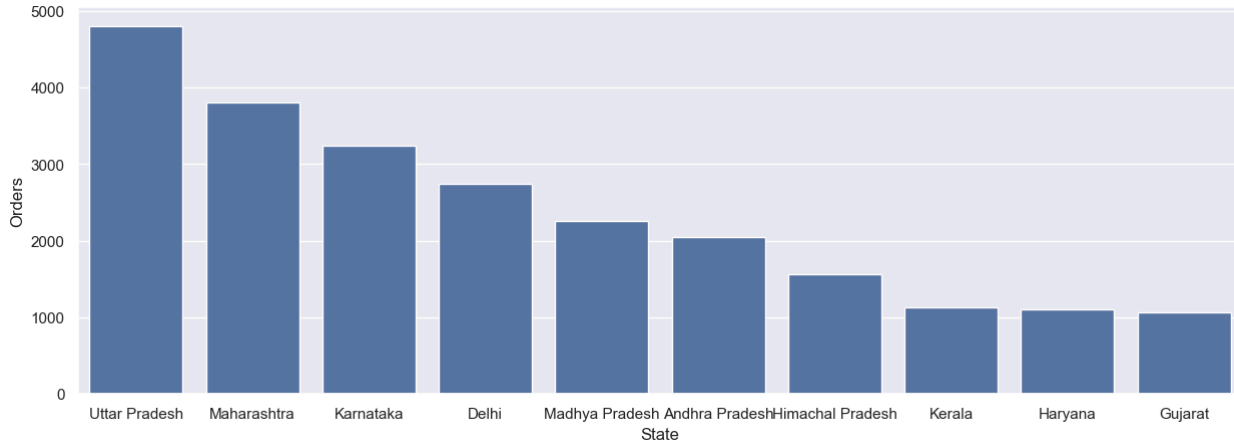
> # Result of bivariant analysis
>
> # From above graph we can conclude that:
>
> > **Age group 26-35 have the most purhasing power**

> # Grouping of Orders by Top 10 States

In [17]:
```
#total amount from states
SalesAreas=df.groupby(['State'],as_index=False)['Orders'].sum().sort_values(by

sns.set(rc={'figure.figsize':(15,5)})
#above code of line is not important it's only used to space graph
sns.barplot(x='State',y='Orders',data=SalesAreas)
```

Out[17]:
```
<Axes: xlabel='State', ylabel='Orders'>
```

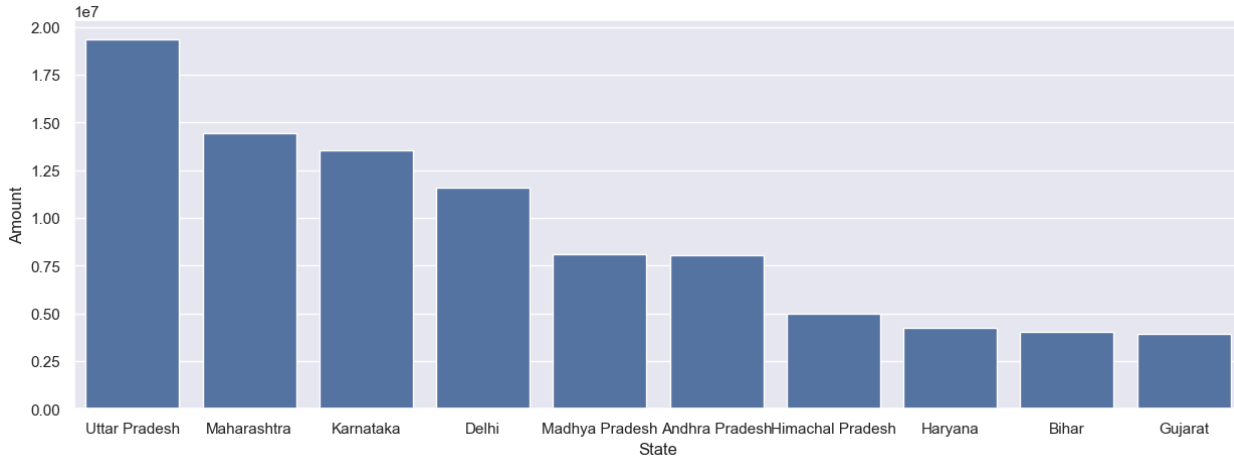


> # Result of bivariant analysis:
>
> > Most orders are from **Utter pradesh, Maharashtra, Karnataka and Delhi**

> # Grouping of Amount of Rupees by Top 10 States

In [18]:
```
SalesAreas=df.groupby(['State'],as_index=False)['Amount'].sum().sort_values(by
sns.barplot(x='State',y='Amount',data=SalesAreas)
```

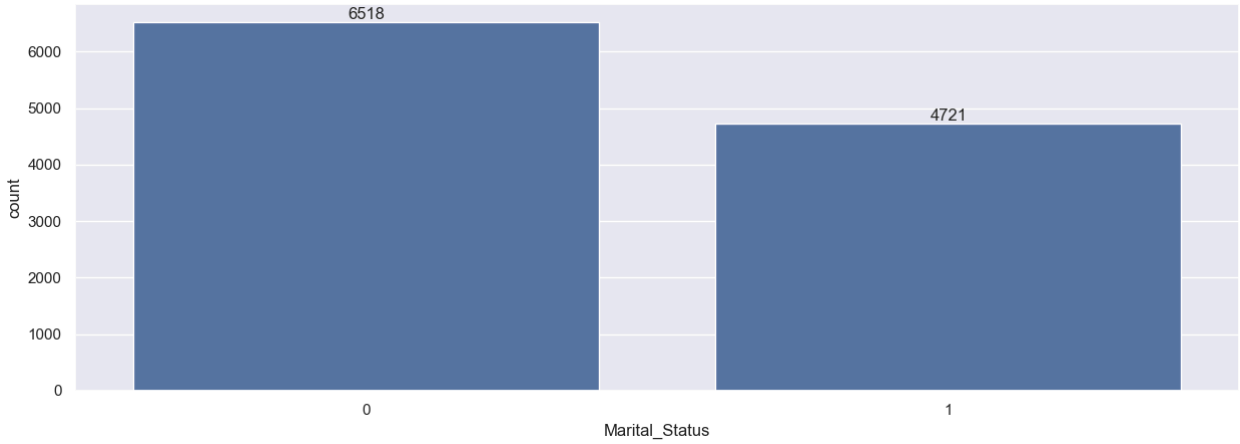
Out[18]: <Axes: xlabel='State', ylabel='Amount'>



> # Result of bivariant analysis:
>
> > Most spending states are also **Utter pradesh, Maharashtra, Karnataka and Delhi**

> counting Customers by maritial status

In [19]:
```
graph=sns.countplot(data=df,x='Marital_Status')
sns.set(rc={'figure.figsize':(3,5)})
for counts in graph.containers:
    graph.bar_label(counts)
```



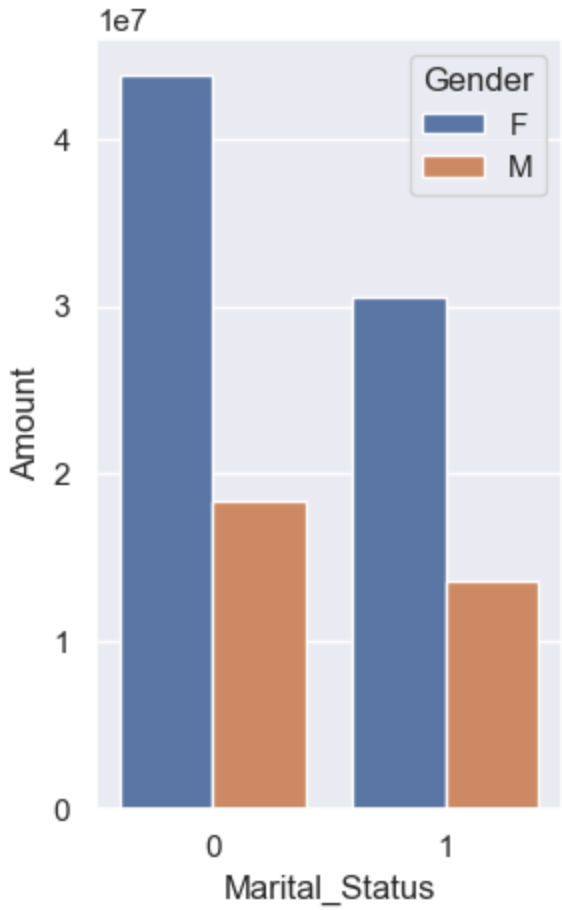
> Result of univariant analysis:
>
> > 57% of our customers are **single**

> Grouping sale amount by marital status and gender

In [20]:
```
saleByMaritalState=df.groupby(['Marital_Status','Gender'],as_index=False)['Amo
```

In [21]:
```
sns.barplot(data=saleByMaritalState,x='Marital_Status',y='Amount',hue='Gender'
```

Out[21]:
```
<Axes: xlabel='Marital_Status', ylabel='Amount'>
```



Result of Multivariant:

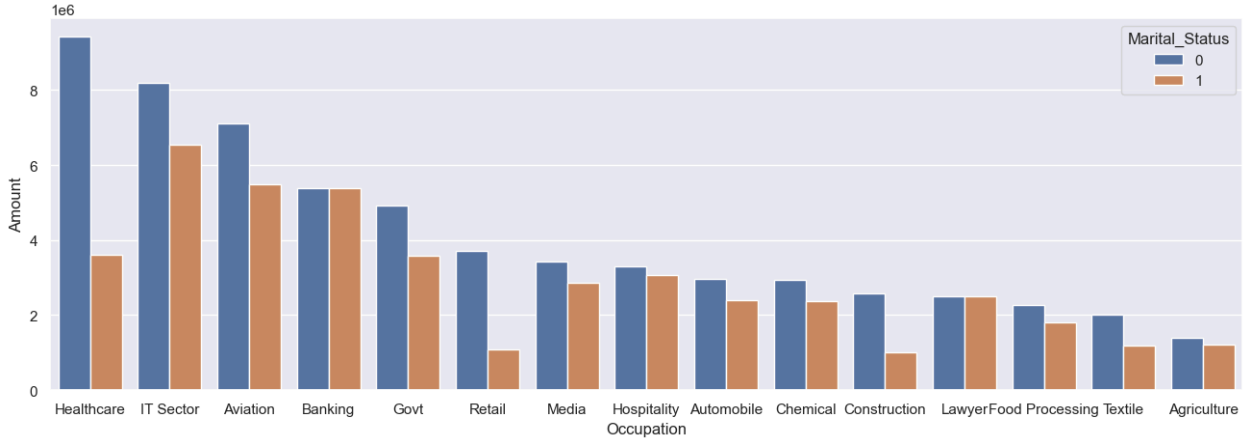
> **Single females are the most spending group**

Graph 1;

> Grouping sale amount by Occpation and Marital status.
>
> # Graph 2;
>
> and Grouping Order Qty by Occpation and Gender.

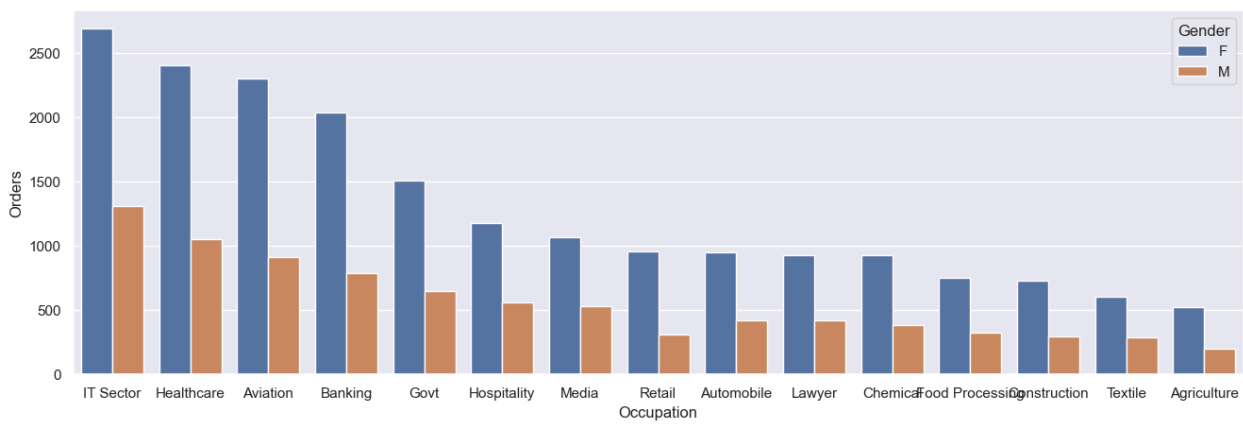
In [22]:
```
#saleAmount by occupation and marital status
saleAmountByOccNgender=df.groupby(['Occupation','Marital_Status'],as_index=Fal
sns.set(rc={'figure.figsize':(16,5)})
sns.barplot(data=saleAmountByOccNgender,x='Occupation',y='Amount',hue='Marital
```

Out[22]: <Axes: xlabel='Occupation', ylabel='Amount'>



In [23]:
```
#OrdersQty by occupation and gender
ordersByOccNgender=df.groupby(['Occupation','Gender'],as_index=False)['Orders'
sns.barplot(data=ordersByOccNgender,x='Occupation',y='Orders',hue='Gender')
```

Out[23]: <Axes: xlabel='Occupation', ylabel='Orders'>



Result of 2 graphs multivariant analysis:

Graph 1:

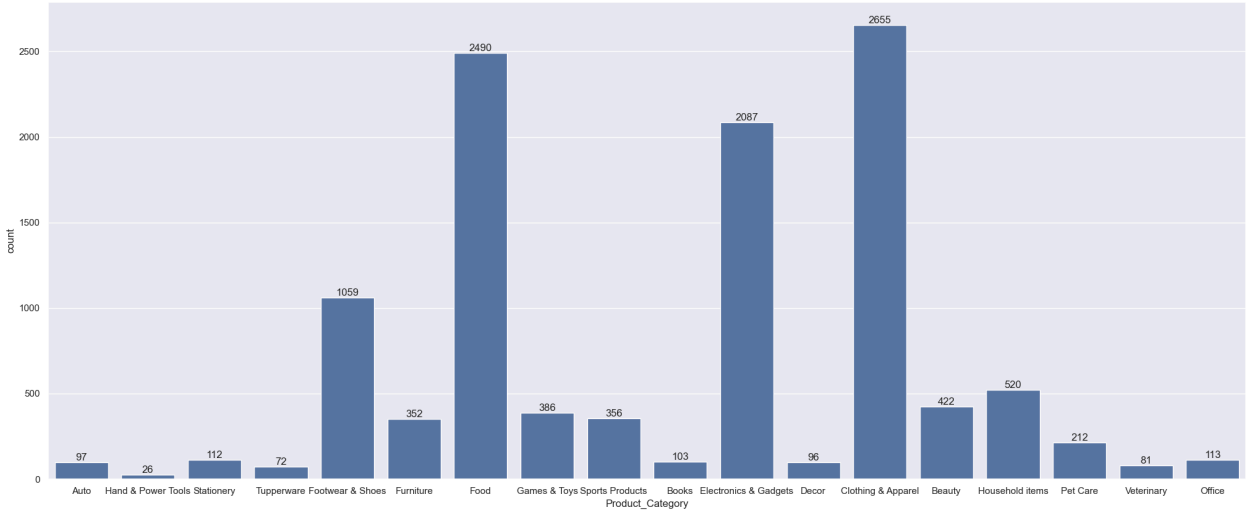
> Sale Amount is dominated by singles working in IT Sector, Healthcare and Avation

Graph 2:

> Quantity is dominated by females working in IT Sector, Healthcare and Avation

analyzing product categories

In [24]:
```
sns.set(rc={'figure.figsize':(25,10)})
graph=sns.countplot(data=df,x='Product_Category')
for count in graph.containers:
    graph.bar_label(count)
```



Result of Univariant analysis

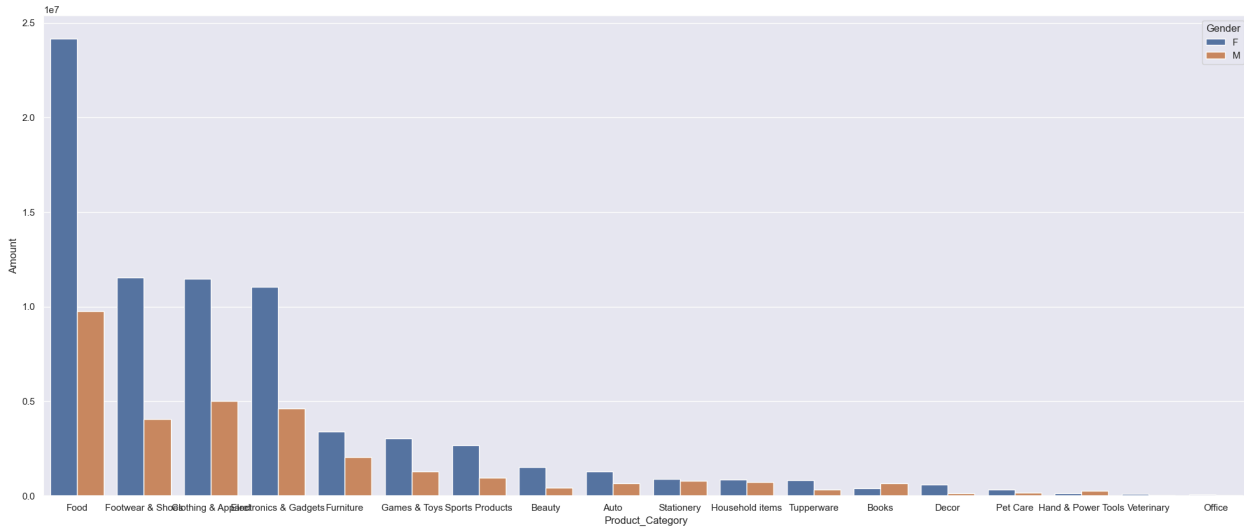
> Most bought product categories are **Clothing & Apparel, Food, Electronic & Gadgets and Footwear & Shoes**

ONLY FOR ADDITIONAL DATA AND STATISTICS

> Analysis of SaleAmount by ProdCategories, Gender, MartStatus
>
> Analysis of OrderQty by ProdCategories, Gender, MartStatus

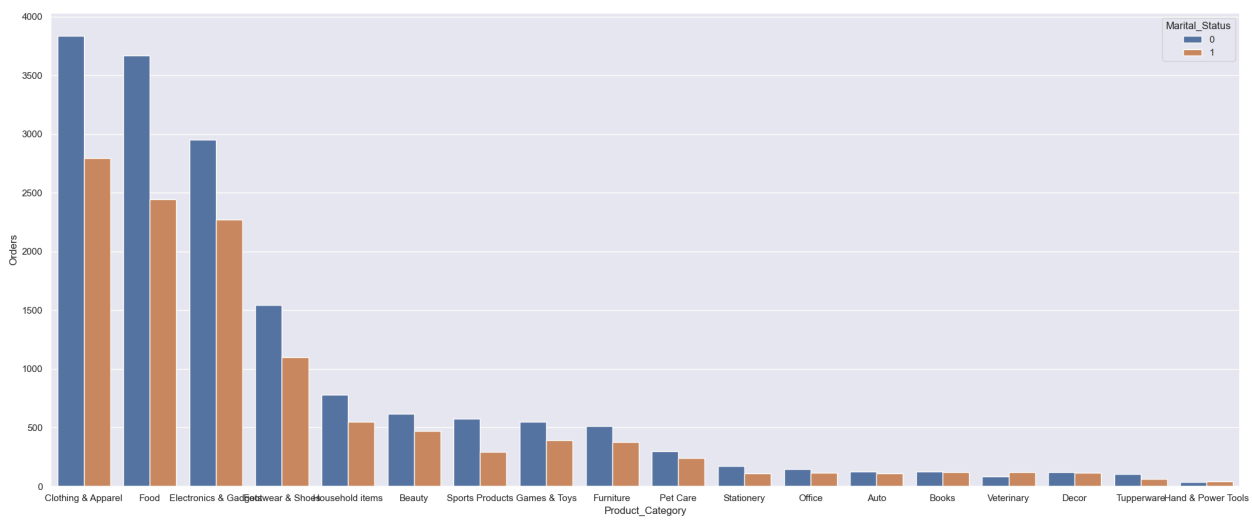
In [25]:
```
saleAmoprod=df.groupby(['Product_Category','Gender',],as_index=False)['Amount'
sns.barplot(data=saleAmoprod,x='Product_Category',y='Amount',hue='Gender')
```

Out[25]: <Axes: xlabel='Product_Category', ylabel='Amount'>



In [32]:
```
saleQtyProd=df.groupby(['Product_Category','Marital_Status'],as_index=False)['
sns.barplot(data=saleQtyProd,x='Product_Category',y='Orders',hue='Marital_Stat
```

Out[32]: <Axes: xlabel='Product_Category', ylabel='Orders'>

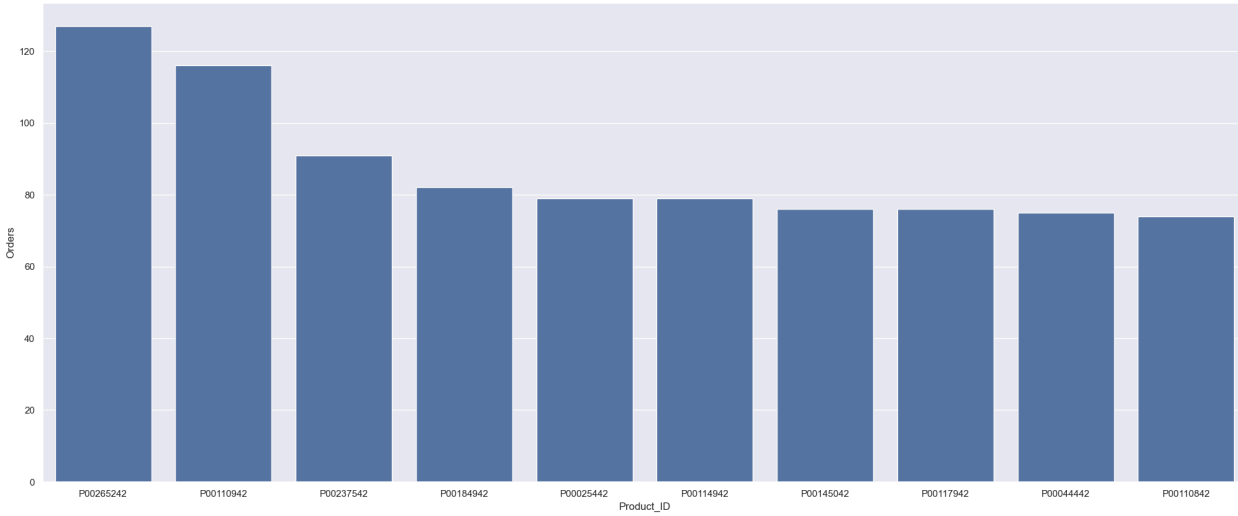


Top 10 most bought products IDs

In [27]:
```
saleItem=df.groupby(['Product_ID'],as_index=False)['Orders'].sum().sort_values
```

```
sns.barplot(data=saleItem,x='Product_ID',y='Orders')
```

Out[27]: <Axes: xlabel='Product_ID', ylabel='Orders'>



EDA Steps 3, 4 are missing because there was no need to Identify outliers and Transform data.

> - No outliers found
> - Data Already Structured

Step 6 Conclusion / Insights

> 1:Data imported
>
> step 2..EDA 1 :Statistical data analysis
>
> 3:Data cleaned
>
> > - Deleted missing columns
> > - Removed missing values
>
> step 4..EDA 2 :Data Visualization
>
> > - Perfomed univariant analysis
> > - Performed bivariant analysis
> > - performed multivariant analysis

Step 6 Conclusion / Insights

The business lacked a granular understanding of its primary revenue drivers.

Generic marketing strategies were resulting in suboptimal engagement, as the company could not clearly distinguish between high-value customer segments and low-interaction groups.

By mapping the intersection of purchasing power and demographic profile, I developed a prioritized customer persona (Single females, 26–35, working in IT/ Healthcare/Aviation). This allows the business to transition from mass-market advertising to targeted, high-conversion campaigns that specifically address the preferences of this high-value cohort.

> Core Data Insights
>
> > - High-Value Demographic: The 26–35 age bracket holds the highest purchasing power marketing ROI will be maximized by focusing exclusively on this segment.
> > - Professional Alignment: The concentration of sales in the IT, Healthcare, and Aviation
> > - sectors suggests a need for industry-specific promotional content.
> > - Geographic Concentration: With major activity in UP, Maharashtra, Karnataka, and Delhi, logistics and regional ad spend should be optimized to these specific high-density hubs.
> > - The "Single" Factor: Since 57% of customers are single and represent the highest spenders, loyalty programs and product bundles should be tailored to individual-use cases rather than household/family-centric consumption.
> > - Category Synergy: Cross-promoting Electronics and Apparel will capture the largest share of wallet from your primary target audience.